

Education

- Norwegian University of Science and Technology (NTNU), Trondheim
 - Cybernetics and Robotics - Master's Program (5-year), 2022 - 2027

Work Experience and Technical Experience

- WaterLinked project engineer - team leader - in R&D summer 2026
 - Proposed and lead an initiative to demonstrate target tracking on the Sonar 3D-15.
 - Proposed the implemented design of the target tracking system to perform real time tracking of multiple ROVs in a cluttered environment.
 - Created a proof of concept of velocity and position estimation of an ROV using a mounted Sonar 3D-15 using the tracker to track multiple stationary targets using Python.
- WaterLinked project engineer in R&D summer 2025
 - SLAM using 3D-sonar and DVL on various industrial underwater ROVs. My contribution was that I designed a 3D-SLAM front-end, and parts of the back-end
 - Finished the ROS-2 driver for the Sonar 3D-15
 - Technical writing of 3D-sonar documentation
- Technical Position in Vortex NTNU Software, Perception Team Member, 2024 to 2026
 - Year 1: Developed a ROV-vision pipeline for detecting the 3D-position and rotation of the gripping point of an underwater valve. Using YOLOv8 and geometrical filtering of a depth-image from a stereo camera using classic robot vision techniques like RANSAC and canny edge detection. I mainly used C++ and OpenCV.
 - Year 2: Developed a 2D-SLAM back-end using iSAM2 for a multi-beam 2D sonar system in ROS 2, validated via Stonefish simulation.
- Omega Workshop, Board Member, from 2022 to 2025
- Omega Workshop IT Committee: 2022 to 2025

Academical interests and experience

- The fields of study that interest me the most:
 - Perception including computer and robotic vision and sensor fusion
 - Robust/safe navigation and control
 - Modeling and simulation
 - Nonlinear systems
 - Reinforcement-learning
- Some experience from solo projects
 - 5 legged robot walking gait using reinforcement learning:
<https://github.com/oahelmer/FiveLeggedRobotRLwalk>
 - 2D-boat simulator controlled by reinforcement learning and LOS guidance:
<https://github.com/oahelmer/2d-boatsim>
 - Quadcopter simulation using MPC control: <https://github.com/oahelmer/Quadcopter-Simulation>

Languages

- Norwegian (native)
- English (fluent)